

- 2 (a) The equation of state for an ideal gas may be expressed as

$$pV = NkT.$$

- (i) State the meaning of each of the symbols in this equation.

p :

V :

N :

k :

T : [3]

- (ii) Using the equation of state, derive an expression for the average translational kinetic energy E_K of a particle in the gas in terms of some or all of N , k and T .

$$E_K = [2]$$

